

Pol van Hoften

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Employment

- 2025- **Associate professor** *Zhejiang University*.
- 2023-2025 **Assistant professor (tenured since 2024)** *Vrije Universiteit Amsterdam*.
- 2021-2023 **Postdoctoral fellow** *Stanford University*.

Education

- 2017-2021 **PhD in mathematics**, *King's College London*, supervised by James Newton and Ana Caraiani, Thesis: "Mod p points on Shimura varieties of parahoric level".
- 2012-2017 **Bachelor and Master in mathematics**, *Utrecht University*, *cum laude*.

Grants

- 2024-2028 **NWO Veni €317 221**, Zeta functions of Shimura varieties.

Publications and preprints

- 2026 [A geometric Jacquet–Langlands correspondence for Shimura varieties](#), with Jack Sempliner.
- 2026 [p-adic Maass–Shimura operators on \$\mu\$ -ordinary Igusa varieties](#), with Andrew Graham and Sean Howe, submitted.
- 2026 [Igusa stacks and the cohomology of Shimura varieties II](#), with Patrick Daniels, Dongryul Kim and Mingjia Zhang, submitted.
- 2025 [p-adic Fourier theory in families](#), with Andrew Graham and Sean Howe, submitted.
- 2024 [Igusa stacks and the cohomology of Shimura varieties](#), with Patrick Daniels, Dongryul Kim and Mingjia Zhang, submitted.
- 2024 [On a conjecture of Pappas and Rapoport](#), with Patrick Daniels, Dongryul Kim and Mingjia Zhang, *Mathematische Annalen* **95** (2026), no. 2, 31.
- 2024 [On the Piatetski-Shapiro construction for integral models of Shimura varieties](#), with Jack Sempliner, submitted.
- 2022 [Hecke orbits on Shimura varieties of Hodge type](#), with Marco D'Addezio, submitted.
- 2021 [On the ordinary Hecke orbit conjecture](#), *Algebra and Number theory* **18** (2024), No. 5, 847–898.
- 2021 [Monodromy and irreducibility of Igusa varieties](#), with Luciena Xiao Xiao, *American Journal of Mathematics* **147** (2025), no. 2, 355–400.
- 2020 [Mod \$p\$ points on Shimura varieties of parahoric level](#), with an appendix by Rong Zhou, *Forum of Mathematics, Pi* **12** (2024).

- 2019 [A geometric Jacquet-Langlands correspondence for paramodular Siegel threefolds](#), Mathematische Zeitschrift **299** (2021), 2029–2061 .

Invited lecture series

- Jun 2025 *Mini-course (4.5 hours)*, Workshop "Alpbach 2025".
Jan 2024 *Mini-course (6 hours)*, Yau Mathematical Sciences Center, Tsinghua University and Morningside Center of Mathematics, Chinese Academy of Sciences.

Invited conference talks

- Oct 2026 *Workshop on p -adic Geometry*, Fudan University.
Jul 2026 *Workshop on p -adic Modular Forms and Fontaine–Mazur*, (two talks), Hefei.
Jul 2026 *Sino-French Symposium in Number theory*, Kunming.
Jun 2026 *Categorical Langlands Summit*, (two expository talks), KIAS, Seoul.
Dec 2025 *2025 Taipei conference on arithmetic geometry*.
Nov 2025 *Arithmetic geometry in Shenzhen*.
July 2025 *New tools in the local and global Langlands program*, Hangzhou.
Nov 2024 *Arithmetica transalpina*, Milan.
Sep 2024 *Clay research conference: New advances in the Langlands program*, Oxford.
Sep 2024 *Workshop on p -adic geometry*, Chicago.
Nov 2023 *Compositio Mathematica prize symposium*, University of Amsterdam.
Aug 2023 *Conference on global Langlands, Shimura varieties and $shtukas$* , HIM Bonn.
Feb 2023 *Oberwolfach workshop on arithmetic of Shimura varieties*.
Jun 2022 *Arithmetic and topology around the Langlands program*, Stockholm.
Aug 2021 *Arithmetic of Shimura varieties over global fields*, Cetraro.
Jan 2021 *Galois representations and related topics*, RIMS, Kyoto.
Oct 2019 *Supersingular abelian varieties and related arithmetic*, Nagoya.

Selected invited seminar talks

- Jan 2026 *Number theory seminar*, Oxford University.
Aug 2025 *Seminar*, Hong Kong University.
Jun 2025 *Algebra and number theory seminar*, Vienna.
May 2025 *Séminaire arithmétique et géométrie algébrique*, IRMA Strasbourg.
Apr 2025 *Number theory seminar*, Stanford University.
Mar 2025 *Arithmetic geometry seminar*, Princeton University.
Nov 2024 *Oberseminar*, MPI Bonn.
Nov 2024 *Seminar on Arithmetic Geometry*, TU Darmstadt.
Oct 2024 *Algebra seminar*, Groningen.
Jun 2024 *Arithmetic algebraic geometry seminar*, Münster.
Jun 2024 *Séminaire groupes réductifs et formes automorphes*, Paris.

Mar 2024 *Intercity number theory seminar*, Amsterdam.
 Jan 2024 *Number theory seminar*, Peking University.
 Jan 2024 *Number theory seminar*, Tsinghua University.
 Dec 2023 *Friday arithmetic geometry seminar*, Imperial College London.
 Dec 2023 *London number theory seminar*.
 Apr 2023 *Princeton university/IAS number theory seminar*.
 Feb 2023 *Number theory seminar*, Johns Hopkins University.
 Dec 2022 *Algebra-number theory seminar*, University of Maryland.
 Oct 2022 *Stanford number theory seminar*, Stanford University.
 Sep 2022 *Number theory and representation theory seminar*, University of Utah.
 Jul 2022 *Séminaire arithmétique et géométrie algébrique*, Université Paris-Saclay.
 Jul 2022 *Algebraic geometry seminar*, TU Darmstadt.
 Jun 2022 *Geometry and algebra seminar*, Utrecht University (two talks).
 Jun 2022 *Arithmetic and algebraic geometry seminar*, University of Amsterdam (two talks).
 Jun 2022 *Number theory seminar*, Imperial College London.
 Jun 2022 *Number theory seminar*, University of Cambridge.
 Feb 2022 *Number theory seminar*, University of Wisconsin-Madison.
 Feb 2022 *Number theory seminar*, University of Chicago.
 Oct 2021 *Berkeley-Caltech-Stanford number theory seminar*.
 Jul 2021 *Oberseminar*, University of Duisburg-Essen.
 May 2021 *Number theory seminar*, University of Münster.
 Apr 2021 *Number theory seminar*, Massachusetts Institute of Technology.
 Apr 2021 *RGR webinar*, Institut de Mathématiques de Marseille.
 Mar 2021 *Number theory seminar*, UCLA.
 Feb 2021 *Algebra-number theory seminar*, University of Maryland (two talks).
 Nov 2020 *London-Paris number theory seminar*.
 Nov 2020 *Algebra and geometry seminar*, Stockholm.
 Nov 2020 *Number theory lunch seminar*, University of Sheffield.
 Jun 2020 *Séminaires de géométrie arithmétique et motivique*, Paris 13.
 Apr 2020 *Number theory seminar*, University of Warwick.
 Dec 2019 *Algebra, geometry and number theory seminar*, Leiden University.
 Nov 2019 *ENS de Lyon number theory seminar*, École normale supérieure de Lyon.

Supervision

PhD Students

2026-2030 *Zhu YINUO*.
 2024-2026 *Dongryul Kim (secondary supervisor, primary supervisor Richard Taylor)*.

Master Students

2023-2024 *Pedro Gómez de Terreros Oramas*.

Undergraduate students

- 2025-2026 *Luyan Zhao and Ziang Zhong.*
2024-2025 *Benedek Biczó, David Bukva, Rikkie Gieler.*

Teaching Experience

- 2025- *Zhejiang University.*
○ Algebraic Geometry 2 (Fall 2026)
○ Algebraic Geometry 1 (Spring 2026)
- 2023-2025 *VU Amsterdam.*
○ Linear algebra for FAR/SBI (Spring 2025)
○ Linear algebra for computer science (Spring 2024/Spring 2025)
○ Fourier Analysis (Fall 2023/Fall 2024)
- 2022-2023 *Stanford University.*
○ Topics in algebraic geometry: Deligne–Lusztig theory (Winter 2023)
○ Graduate algebraic geometry (Fall 2022)
○ Reading course on Shimura varieties in characteristic p (Spring 2022)
- 2018-2020 **Graduate teaching assistant** *King's College London.*
○ Groups and Symmetry (Fall 2020)
○ Introduction to number theory (Fall 2019/Fall 2018)
- Summer 2018 **Counselor** *Ross mathematics program Asia*, 5 weeks.
- 2013-2017 **Teaching assistant** *Utrecht University.*
I was a teaching assistant for 7 courses.

Teaching training

- 2023-2024 *University teaching qualification (BKO)*, VU Amsterdam, 6 months.
Fall 2022 *Math teaching workshop*, Stanford University, 3 days.
Fall 2018 *Graduate teaching assistant workshop*, King's College London, 1 day.

Service and other professional activities

- Refereeing *I have refereed ≥ 12 papers..*
- 2024-2025 *Member of the exam board for mathematics/business analytics at VU Amsterdam..*
- 2023 *Member of Sean Cotner's thesis committee.*
- 2022 *Member of Konstantin Miagkov's area exam committee.*

Conference Organization

- 2026 *Arithmetic Algebraic Geometry in Hangzhou*, with A. Mihatsch and Yousheng Shi.

Seminars organised

- Spring 2026 *Condensed mathematics learning seminar*, with Yousheng Shi.
- Fall 2024 *Learning seminar on prismatic cohomology*, with Dhyam Aranha.
- 2022-2023 *Berkeley–Stanford number theory learning seminar on EG stacks*, with Lie Qian.
- Spring 2022 *Berkeley–Stanford number theory learning seminar on moduli of L -parameters.*
- Fall 2021 *Reading group on "A unipotent circle action on p -adic modular forms".*
- Winter 2020 *Reading group on Deligne–Lusztig theory*, with Miriam Norris.

Winter 2019 *Organised the London number theory reading group on 'derived structures in the Langlands programme', with Carl Wang-Erickson, Ashwin Iyengar and Alice Pozzi.*